

SysLifeChecker

Technical Internals: How Health Scores Are Calculated

This document explains, in full technical detail, exactly how SysLifeChecker computes the health percentages and status labels shown in the Dashboard and Report -- battery health, SSD/HDD (SMART-based) health, and the other validation checks -- including the underlying Windows APIs, WMI classes, IOCTLs, and SMART attributes each figure is derived from.

Application: SysLifeChecker

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Platform: Windows 10 / 11 (64-bit)

Data sources: WMI (Win32_*), Win32 Device I/O (IOCTL), smartmontools (smartctl)

Contents

- 1. Overview of Health Scoring Philosophy
- 2. Battery Health Calculation
- 3. SSD / NVMe Health Calculation (SMART-Based)
- 4. HDD Health Calculation (SMART-Based)
- 5. Storage Health Status Labels & Recommendations
- 6. Estimated Remaining Lifetime
- 7. System Validation Checks (Security Card)
- 8. Data Collection Sources Reference
- 9. Known Limitations & Accuracy Notes

1. Overview of Health Scoring Philosophy

SysLifeChecker deliberately uses simple, transparent, threshold-based formulas rather than opaque proprietary scoring. Every health percentage shown in the app is traceable to a small number of real hardware counters -- there is no machine learning model, no hidden weighting, and no cloud lookup involved. This document exists so that the exact formula behind any number in the app can be verified against the source code (src/Battery.cpp and src/Storage.cpp).

All health percentages are clamped to the 0-100 range and mapped to one consistent three-tier verdict system used across the whole app:

Range	Verdict	Color	Meaning
≥ 50%	Healthy	Green	Operating normally, no action needed
30% - 49%	Attention / Fair	Amber	Showing wear; monitor, no urgent action
< 30%	Critical / Replace	Red	Significant degradation; action recommended

Note: the Storage/SMART subsystem additionally uses a finer-grained four-tier label (Healthy / Good / Warning / Poor) for its human-readable recommendation text -- see Section 5.

2. Battery Health Calculation

2.1 Data source

Battery data is read from two places, in priority order: (1) Win32_Battery via WMI for identity and design/full-charge capacity in mWh, cross-checked against (2) the Windows Battery IOCTL API (IOCTL_BATTERY_QUERY_INFORMATION / IOCTL_BATTERY_QUERY_STATUS via a handle opened with SetupDiGetClassDevs/IOCTL_BATTERY_QUERY_TAG), which gives the most accurate current capacity and cycle count. GetSystemPowerStatus() is used only as a last-resort fallback if no battery device is found via WMI/IOCTL at all (e.g. some laptops with ACPI control-method batteries).

2.2 Health percentage formula

Battery health is a wear metric: how much of the battery's original (design) capacity it can still hold at full charge.

$$\text{HealthPercent} = (\text{FullChargeCapacity_mWh} / \text{DesignCapacity_mWh}) \times 100$$

If either value is unavailable (DesignCapacity or FullChargeCapacity is 0 -- happens on some battery controllers that don't report design capacity), SysLifeChecker does not guess: it defaults HealthPercent to 100 rather than showing a misleading number, since no wear can be measured without a baseline to compare against.

2.3 Healthy / Attention / Replace verdict

- **Healthy** = HealthPercent $\geq$ 50%
- **Attention (Fair)** = $30\% \leq$ HealthPercent $<$ 50%
- **ReplaceRecommended** = HealthPercent $<$ 30% OR CycleCount $>$ 800

The cycle-count override exists because a battery can retain high rated capacity while still having degraded internal chemistry from heavy cycling; 800 full-equivalent cycles is used as a practical, conservative threshold for most consumer lithium-ion laptop batteries.

2.4 The "255%" sentinel bug and how it's avoided

The Win32 SYSTEM_POWER_STATUS.BatteryLifePercent field uses the literal byte value 255 to mean "percentage unknown" -- not a real reading. SysLifeChecker explicitly checks for and discards this sentinel rather than displaying it, so the app never shows a nonsensical "255%" battery level:

```
if (status.BatteryLifePercent != 255)
    battery.RemainingCapacityPercent = status.BatteryLifePercent;
```

2.5 Desktop PCs (no battery)

If Windows reports no real battery device -- or only a "phantom" WMI battery entry with zero design capacity, zero full-charge capacity, and zero remaining percent (seen on some desktop motherboards with ACPI quirks) -- SysLifeChecker discards that phantom entry and instead reports "Desktop AC Power" with no battery percentage shown at all, rather than inventing a number.

Phantom-entry filter: a battery record is discarded if DesignCapacitymWh == 0 AND FullChargeCapacitymWh == 0 AND RemainingCapacityPercent == 0 simultaneously.

2.6 Capacity figures shown on the Dashboard

Field	Meaning	Source
Design Capacity (original)	Capacity when the battery was new	WMI DesignCapacity
Full Charge Capacity (now)	Maximum charge it can currently hold	IOCTL_BATTERY_QUERY_INFORMATION
Currently Available	Charge in the battery right now (Wh and %)	IOCTL_BATTERY_QUERY_STATUS
Cycle Count	Number of full charge/discharge cycles	WMI Win32_Battery / IOCTL

3. SSD / NVMe Health Calculation (SMART-Based)

3.1 Data source

All SMART data (SSD, NVMe, and HDD) is read by shelling out to the bundled smartctl tool (from the open-source smartmontools project, `third_party/smartmontools/smartctl.exe`) with JSON output (`smartctl --scan-open -j` to enumerate devices, then `smartctl -a -j <device>` per device), parsed with `nlohmann::json`. Devices reported by smartctl are matched back to the drives already enumerated via `Win32_DiskDrive` by normalized serial number, then normalized model name, then (if only one drive exists) by elimination.

3.2 NVMe/SSD formula

NVMe and SATA SSD health starts from the drive's own self-reported wear-level attribute (`percentage_used` from the NVMe SMART health log, a value the drive controller itself calculates from flash program/erase cycle wear), then applies penalties for hard failure indicators:

```
health = 100 - PercentageUsed
if (MediaErrors > 0): health -= 20
if (SMART overall-health self-assessment == FAILED): health = 0
health = clamp(health, 0, 100)
```

`PercentageUsed` is the NVMe spec's own "percentage of the device's rated endurance consumed" figure, and `PercentageRemaining` shown elsewhere is simply `100 - PercentageUsed`.

4. HDD Health Calculation (SMART-Based)

Traditional spinning hard drives have no single "wear used" counter (no flash endurance), so HDD health instead starts at a perfect 100 and is penalized for specific ATA SMART attributes that are reliable, well-documented predictors of mechanical/media failure:

SMART Attribute (ID)	Meaning	Penalty
Reallocated Sectors Count (5)	Sectors physically remapped after failing	-20 if > 0
Current Pending Sector Count (197)	Sectors awaiting remap (unstable)	-30 if > 0
Offline Uncorrectable (198)	Sectors that failed to read even offline	-40 if > 0
SMART overall-health self-assessment	Drive firmware's own failure prediction	Forces health = 0

```
health = 100
if (ReallocatedSectors > 0): health -= 20
if (PendingSectors > 0): health -= 30
```

```
if (UncorrectableErrors > 0): health -= 40
if (SMART predicts failure): health = 0
health = clamp(health, 0, 100)
```

These three attributes are penalized independently and can stack -- a drive with both reallocated and pending sectors drops by 50 points, reflecting compounding real-world risk.

Other collected attributes -- Power-On Hours (9), Power Cycle Count (4), Spin Retry Count (10), Start/Stop Count (12), Load/Unload Cycles (193), and CRC Errors (199) -- are recorded and shown in the full report for diagnostic context, but do not currently factor into the numeric health score itself.

5. Storage Health Status Labels & Recommendations

The numeric SMART health score (Sections 3-4) is mapped to a four-tier human-readable label, independent of the battery/validation three-tier system, giving finer resolution specifically for storage:

Condition	Label	Recommendation shown to user
SMART predicts imminent failure	Critical	SMART predicts imminent drive failure. Replace immediately.
health $\geq$ 80	Healthy	Drive is operating normally.
50 $\leq$ health < 80	Good	Drive is healthy but showing normal wear.
40 $\leq$ health < 50	Warning	Back up important data and monitor drive health.
health < 40	Poor	Drive health is significantly degraded. Replacement is recommended.

If smartctl cannot be reached at all (missing binary, unsupported controller, or a USB/RAID enclosure that blocks SMART passthrough), SysLifeChecker does not show a false warning -- it defaults that drive to "Healthy" / 100% with the note "Drive is operating normally," since an absence of SMART data is not evidence of a problem.

6. Estimated Remaining Lifetime

SysLifeChecker calculates an internal EstimatedRemainingYears value per drive for diagnostic/report purposes, using one of two methods depending on drive type:

For SSD/NVMe (wear-based): extrapolates linearly from age and wear used so far --

```
EstimatedAgeYears = PowerOnHours / 24 / 365.25
EstimatedRemainingYears = (EstimatedAgeYears / PercentageUsed) × (100 - PercentageUsed)
```

For HDD or when wear data is unavailable: falls back to a generic manufacturer-rated operating-hours budget of 50,000 hours, subtracting hours already logged:

```
remainingHours = max(50000 - PowerOnHours, 8760) // floor of 1 year
EstimatedRemainingYears = remainingHours / (24 × 365.25)
```

This estimate was removed from the Dashboard and Report UI at the app owner's request, since a simple linear extrapolation is a rough approximation, not a reliable failure prediction -- it remains in the underlying data model for potential future use but is not currently shown to end users anywhere in the app.

7. System Validation Checks (Security Card)

The Dashboard's System Validation card is a straightforward pass/fail readout of security-relevant system state -- there is no scoring formula here, just direct queries:

Check	Source	Pass condition
Secure Boot	WMI / firmware UEFI variables	SecureBootEnabled == true
TPM Security	WMI Win32_Tpm	TPM present, version reported (2.0 = "Ready")
Antivirus Protection	WMI SecurityCenter2 / Windows Defender status	Defender enabled OR a 3rd-party AV registered
Windows Firewall	WMI firewall profile status	Firewall.Enabled == true
Windows License	WMI SoftwareLicensingProduct	Activation.Activated == true
Drive Encryption	WMI Win32_EncryptableVolume (BitLocker)	At least one drive reports Encrypted == true

8. Data Collection Sources Reference

Component	Primary API / Source
CPU	WMI Win32_Processor, registry, CPUID
RAM	GlobalMemoryStatusEx, WMI Win32_PhysicalMemory, GetPerformanceInfo (commit/pagefile)
GPU	WMI Win32_VideoController, DXGI
Storage (identity)	WMI Win32_DiskDrive, Win32_DiskPartition, GetDiskFreeSpaceEx
Storage (SMART/health)	smartmontools smartctl (JSON mode), bundled in third_party/
Storage (features)	IOCTL_STORAGE_QUERY_PROPERTY, IOCTL_DISK_GET_DRIVE_LAYOUT_EX
Battery	WMI Win32_Battery, IOCTL_BATTERY_QUERY_INFORMATION/STATUS, GetSystemPowerStatus (fallback)
Network	WMI Win32_NetworkAdapter / Win32_NetworkAdapterConfiguration, WLAN API
Motherboard / BIOS	WMI Win32_BaseBoard, Win32_BIOS
Windows / Activation	WMI Win32_OperatingSystem, SoftwareLicensingProduct
Security	WMI SecurityCenter2, Win32_Tpm, Win32_EncryptableVolume, firewall profile APIs
Audio / Cameras	WMI Win32_SoundDevice, Media Foundation device enumeration
Drivers	SetupAPI device/driver enumeration

9. Known Limitations & Accuracy Notes

- **SMART data availability depends on the storage controller/enclosure** -- USB and some RAID controllers block SMART passthrough entirely, in which case SysLifeChecker defaults to "Healthy" rather than guessing (see Section 5).
- **Battery design capacity is not reported by every battery controller**; when unavailable, health defaults to 100% since wear cannot be measured without a baseline.

- **The remaining-lifetime estimate (Section 6) is a simple linear extrapolation**, not a statistical failure model, and is intentionally hidden from the UI for this reason.
- **The HDD SMART penalty weights (-20/-30/-40) are fixed**, judgment-based thresholds chosen for clarity and consistency, not a manufacturer-certified scoring standard -- treat the resulting percentage as a strong indicator, not a certified diagnostic.
- **All thresholds and formulas in this document reflect the current source code at the time of writing** and may be refined in future updates; the source files (src/Battery.cpp, src/Storage.cpp) are always the ground truth.